25901648 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATIONS, APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCCFS101 - FUNDAMENTALS OF BLOCK CHAIN AND CRYPTOCURRENCY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Distinguish between two main types of faults in distributed system? [U] / [CO1]
2. Discuss about decentralized consensus. [U] / [CO1]
3. Explain PBFT. [U] / [CO1]
4. Explain any 4 benefits of using decentralization. [U] / [CO1]
5. Outline the challenges associated with the Proof of Work (PoW) consensus model. [An] / [CO2]
6. Examine the purpose of Bitcoin Improvement Proposals (BIPs) in the blockchain ecosystem. [An] / [CO3]
7. Define and examine asymmetric cryptography. [An] / [CO2]
8. Explain confidentiality in the context of cryptography. [U] / [CO1]
9. Rank Bitcoin payment methods based on transaction speed and cost. [E] / [CO3]

10. Conclude how Bitcoin's Proof of Work (PoW) consensus mechanism secures the network. [E] / [CO3]

11. Rate the usability of mobile wallets compared to desktop wallets. [E] / [CO3]

12. Conclude why Bitcoin transactions are irreversible. [E] / [CO3]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

13. Discuss the importance of consensus in blockchain technology. [U] / [CO1]

14. Explain why mesh networks are important for decentralized communication. [U] / [CO1]

15. Judge the risks involved in buying Bitcoin from unregulated exchanges. [E] / [CO3]

16. Explain block ciphers and its working with a neat diagram. [U] / [CO1]

[5x3 = 15]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

17. Explain the structure and working of blockchain with the help of diagram. [U] / [CO1]

18. Select the most reliable software for installing and managing a Bitcoin wallet. [E] / [CO3]

[10x1 = 10]